TCS NQT PYQ QUESTIONS

Company: TCS NQT
Round: Online
Year: 2025
Difficulty: High
Topic: Number Systems (Remainders / CRT)

Problem Description:

A number N, when divided by 7, leaves a remainder of 3. When divided by 11, it leaves a remainder of 7. When divided by 13, it leaves a remainder of 9.

Find the remainder when N is divided by 1001 (note that $1001 = 7 \times 11 \times 13$).

Options:

A) 993
B) 995
C) 997
D) 999

Correct Answer:

C) 997

Approach to Solve This Problem:

STEP 1:

Look for a common shift that makes all three remainders vanish at once. Add 4 to N and check each condition:

$N + 4 \equiv 3 + 4 = 7 \equiv 0 \pmod 7$

$N + 4 \equiv 7 + 4 = 11 \equiv 0 \pmod{11}$

$N + 4 \equiv 9 + 4 = 13 \equiv 0 \pmod{13}$

STEP 2:

So N + 4 is divisible by 7, 11 and 13 individually. Since 7, 11 and 13 are pairwise coprime, N + 4 must be divisible by their product, 1001.

Hence $N + 4 = 1001k$ for some positive integer k, which gives $N = 1001k - 4$.

STEP 3:

Writing this as $N = 1001(k-1) + (1001 - 4) = 1001(k-1) + 997$ shows that dividing N by 1001 leaves a remainder of 997.

CORRECT ANS = 997